Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

WINTER SEMESTER 2025-26

ASSESSMENT - 1

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Registration No: 23MIS0178

School: School of Computer Science Engineering and Information Systems (SCORE)

Course Code: ISWE406P

Course Title: Agile Development Process And Devops Lab

Faculty: Dr. Ramalingam M

Slot: L57+L58

Experiment 1

Develop a simple application (console-based or GUI-based) using any programming language of your choice. Initialize a local Git repository for the project and manage the source code using Git version control. Commit the application source code with proper commit messages and maintain version history.

Create a remote repository on GitHub and push the local repository to GitHub. Verify successful hosting by accessing the repository online.

Perform branch-related operations by creating at least one feature branch from the main branch, implement a minor enhancement or modification in the feature branch, commit the changes, and merge the feature branch back into the main branch. Resolve any merge conflicts if they arise and push the updated main branch to GitHub.

Step 1: Project Creation

A project folder named **SIMPLE_CALCULATOR** was created and opened using **Visual Studio Code**.
A C++ source file named **calculator.cpp** was created and a simple calculator program (addition operation) was written.

Step 2: Git Repository Initialization

The local Git repository was initialized using the following command:

```
git init
```

Step 3: Checking Repository Status

The status of the repository was verified using the command:

```
git status
```

Step 4: Adding Files to Staging Area

The C++ source file was added to the staging area:

```
git add calculator.cpp
```

Step 5: Committing the Source Code

The staged file was committed with an appropriate commit message:

```
git commit -m "Initial commit: Added calculator with addition operation"
```

Step 6: Branch Configuration

The default branch name was changed to main:

“git branch -M main”

Step 7: Creating Remote Repository

A new repository was created on GitHub and linked to the local repository:

“git remote add origin https://github.com/jaswanthhitman45/calculator”

Step 8: Pushing Code to GitHub

The local repository was pushed to GitHub using the command:

git push -u origin main

The successful hosting of the repository was verified by accessing it online.

```
PS C:\Users\Lenovo\Desktop\SimpleCalculator> git init
>>
Initialized empty Git repository in C:/Users/Lenovo/Desktop/SimpleCalculator/.git/
PS C:\Users\Lenovo\Desktop\SimpleCalculator> git status
>>
No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        calculator.cpp

nothing added to commit but untracked files pgit add calculator.cppto track)
>> C:\Users\Lenovo\Desktop\SimpleCalculator>
PS C:\Users\Lenovo\Desktop\SimpleCalculator> git commit -m "Initial commit: Added basic calculator for addition"
>>
[master (root-commit) 69ec4fc] Initial commit: Added basic calculator for addition
 1 file changed, 11 insertions(+)
 create mode 100644 calculator.cpp
PS C:\Users\Lenovo\Desktop\SimpleCalculator> git log
>>
commit 69ec4fc841cd5c636c85b8495b454f289d5c0c1c (HEAD -> master)
Author: POTHURU-JASWANTH <pothuru.jaswanth2023@vitstudent.ac.in>
Date:   Thu Jan 8 18:06:07 2026 +0530

    Initial commit: Added basic calculator for addition
```

```
    Initial commit: Added basic calculator for addition
PS C:\Users\Lenovo\Desktop\SimpleCalculator> git branch -M main
PS C:\Users\Lenovo\Desktop\SimpleCalculator> git remote add origin https://github.com/jaswanthhitman45/calculator.git
PS C:\Users\Lenovo\Desktop\SimpleCalculator> git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 400 bytes | 100.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/jaswanthhitman45/calculator.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Step 9: Feature Branch Creation

A new feature branch was created from the main branch to add an enhancement:

git checkout -b feature-subtraction

Step 10: Enhancement Implementation

In the feature branch, a **subtraction operation** was added to the existing calculator program.

Step 11: Commit Feature Changes

The modified file was staged and committed using the following commands:

git add calculator.cpp

git commit -m "Added subtraction feature to calculator"

```
PS C:\Users\Lenovo\Desktop\SimpleCalculator> git checkout -b feature-subtraction
>>
Switched to a new branch 'feature-subtraction'
PS C:\Users\Lenovo\Desktop\SimpleCalculator> git branch
>>
* feature-subtraction
  main
PS C:\Users\Lenovo\Desktop\SimpleCalculator> git add calculator.cpp
PS C:\Users\Lenovo\Desktop\SimpleCalculator> git commit -m "Added subtraction functionality"
[feature-subtraction f8642d0] Added subtraction functionality
 1 file changed, 1 insertion(+)
PS C:\Users\Lenovo\Desktop\SimpleCalculator> git push origin feature-subtraction
>>
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 369 bytes | 123.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature-subtraction' on GitHub by visiting:
remote:      https://github.com/jaswanthhitman45/calculator/pull/new/feature-subtraction
remote:
To https://github.com/jaswanthhitman45/calculator.git
 * [new branch]      feature-subtraction -> feature-subtraction
```

Step 12: Merging Feature Branch

The branch was switched back to the main branch and the feature branch was merged:

git checkout main

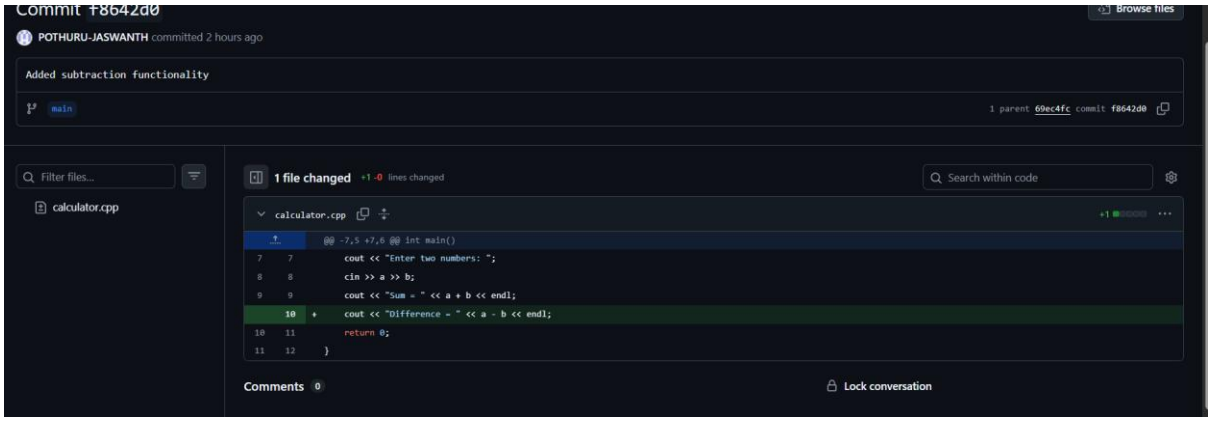
git merge feature-subtraction

Step 13: Final Push

The updated main branch was pushed to GitHub:

git push origin main

```
 * [new branch]      feature-subtraction -> feature-subtraction
PS C:\Users\Lenovo\Desktop\SimpleCalculator> git checkout main
>>
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\Lenovo\Desktop\SimpleCalculator> git merge feature-subtraction
>>
Updating 69ec4fc..f8642d0
Fast-forward
 calculator.cpp | 1 +
 1 file changed, 1 insertion(+)
PS C:\Users\Lenovo\Desktop\SimpleCalculator> git push origin main
>>
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/jaswanthhitman45/calculator.git
   69ec4fc..f8642d0  main -> main
PS C:\Users\Lenovo\Desktop\SimpleCalculator>
```



Experiment 2

Form a team of five members to collaboratively develop a simple application using Gitand GitHub. One member shall act as the repository owner, while the remaining members shall act as collaborators.

Create a shared GitHub repository and clone it to all team members' local systems. Each team member must create a separate feature branch from the main branch and modify the same source file by adding their own functionality or contribution.

Each collaborator shall commit and push their changes to their respective branches and raise a pull request to the main branch. The repository owner shall review the pull requests, resolve any merge conflicts that arise due to simultaneous modification of the same file, and merge the changes into the main branch. Finally, push the updated main branch to GitHub and verify that the final application includes contributions from all five team members. Document the Git commands used and explain the collaborative workflow followed in the experiment.

Team Details

- **Repository Owner:** Marimamula Akshay Guptha
- **Collaborators:**
    1. Jaswanth Pothuru
    2. Dhanush Reddy
    3. Koyahari Krishna
    4. Praneeth Tullala

Step 1: Repository Creation

The repository owner created a GitHub repository named **scm-team-exp** and added collaborators using the *Manage Access* option.

Step 2: Cloning the Repository

The repository was cloned into the local system using Visual Studio Code:

"git clone https://github.com/Akshay-Guptha/scm-team-exp.git"

```
PS C:\Users\Lenovo\Desktop> git clone https://github.com/Akshay-Guptha/scm-team-exp.git
Cloning into 'scm-team-exp'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 1), reused 5 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
Resolving deltas: 100% (1/1), done.
PS C:\Users\Lenovo\Desktop>
```

Step 3: Base Application Creation

The owner created the base Java file **TeamApp.java** and added the initial code.

Step 4: Committing and Pushing Base Code

The base application was staged, committed, and pushed to the main branch:

"git add TeamApp.java"

"git commit -m "Base team application""

"git push -u origin main"

Step 5: Feature Branch Creation by Collaborator

Each collaborator created a separate feature branch from the main branch:

“git checkout -b feature-member1”

Step 6: Contribution by Team Member

The collaborator modified the same file **TeamApp.java** by adding their own contribution message.

Step 7: Commit and Push by Collaborator

The changes were committed and pushed to the respective feature branch:

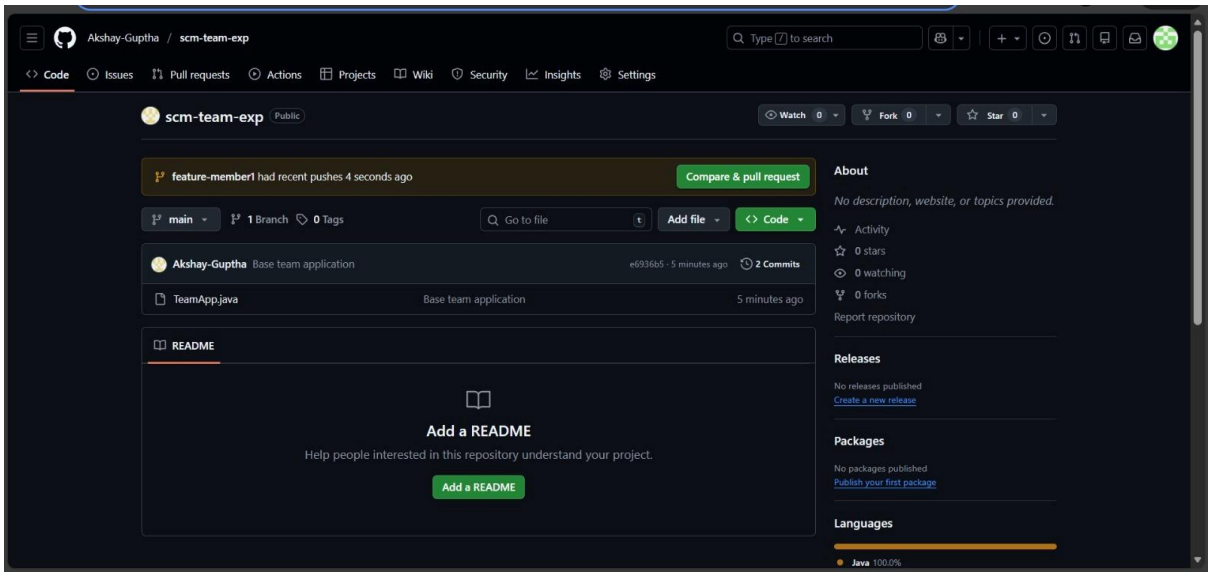
“git add .”

“git commit -m "Added contribution by Member 1"”

“git push origin feature-member1”

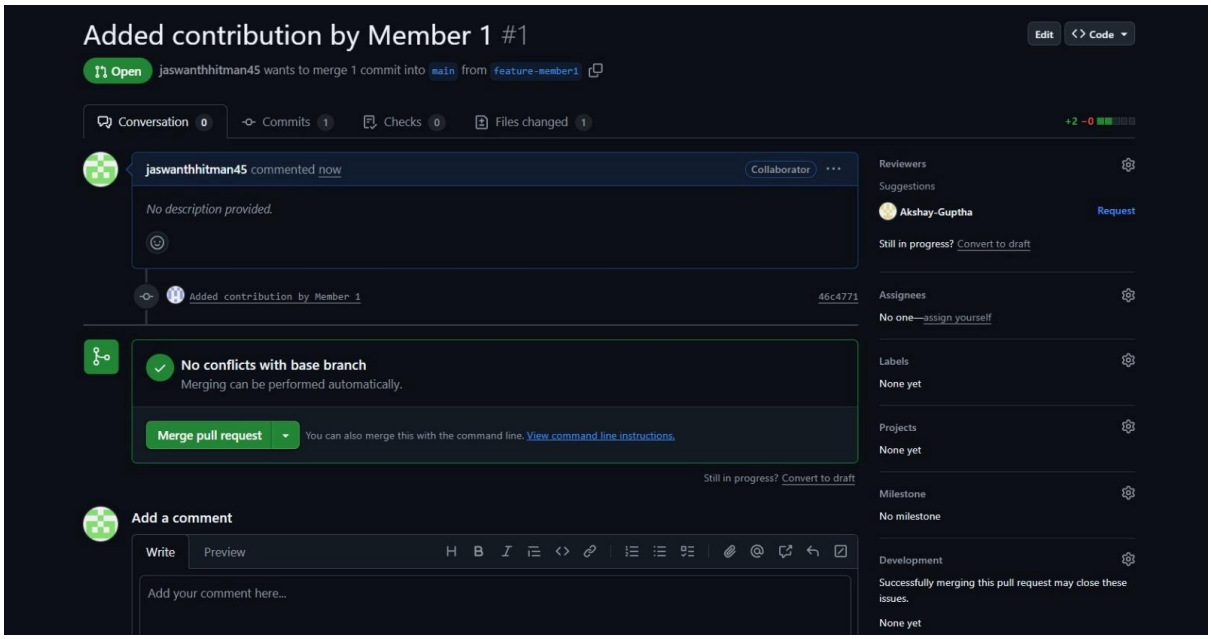
Step 8: Pull Request Creation

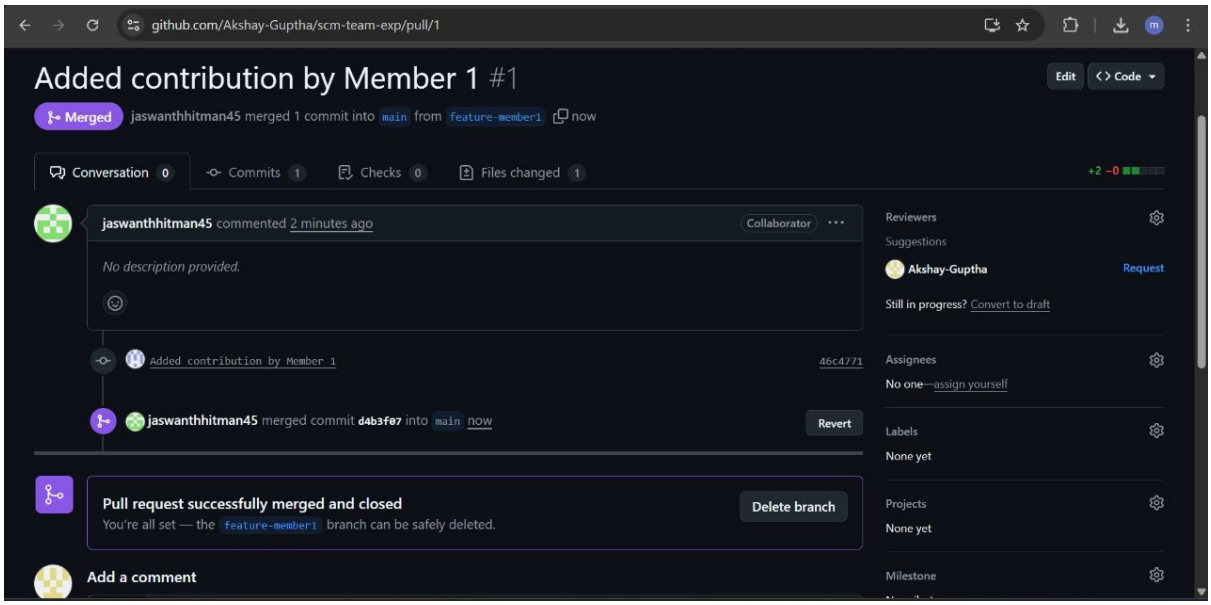
The collaborator raised a pull request from feature-member1 to the main branch using GitHub.

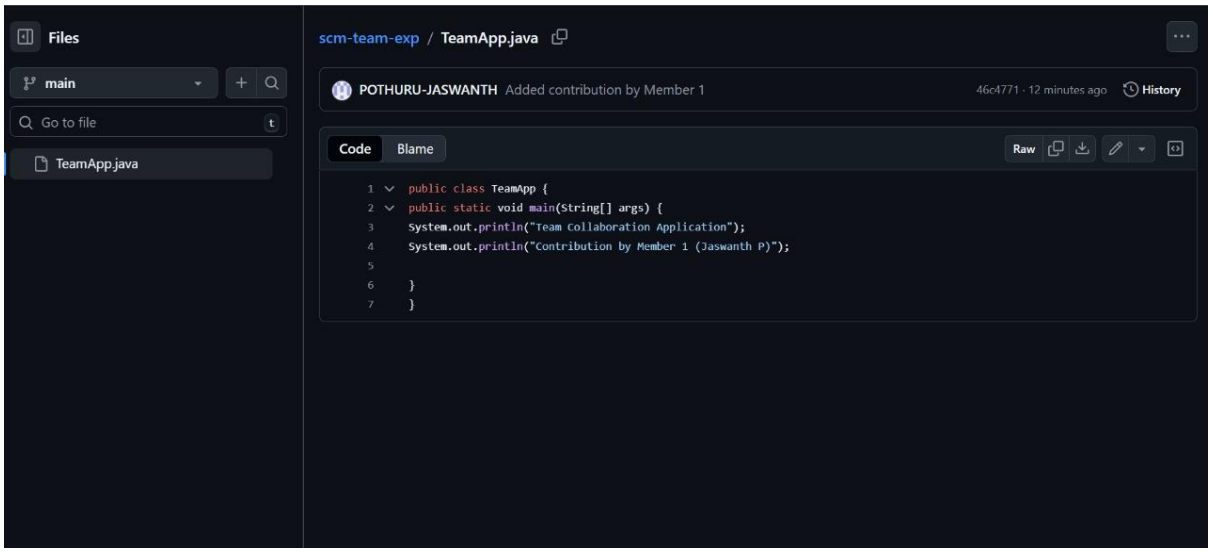
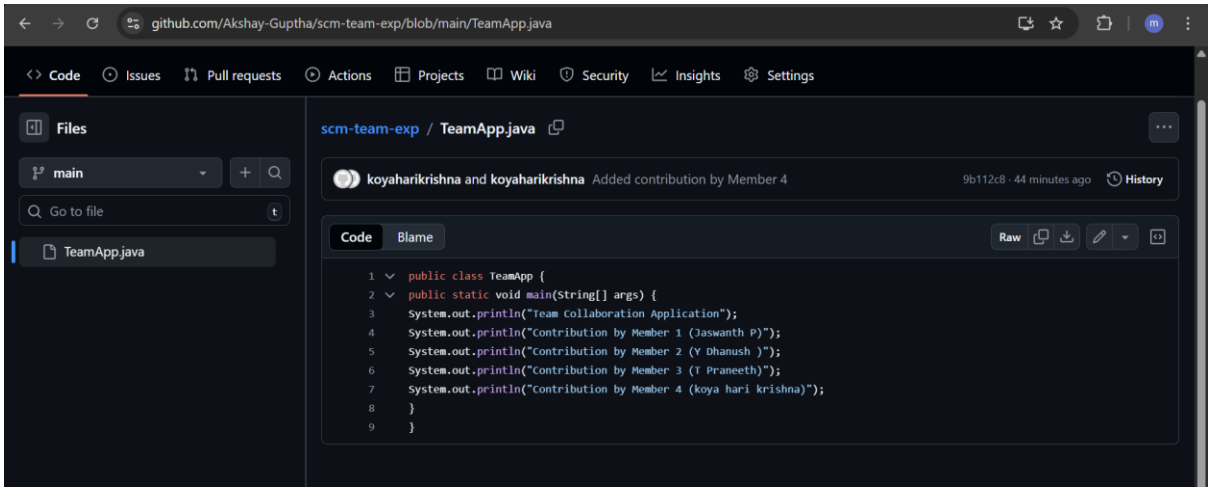
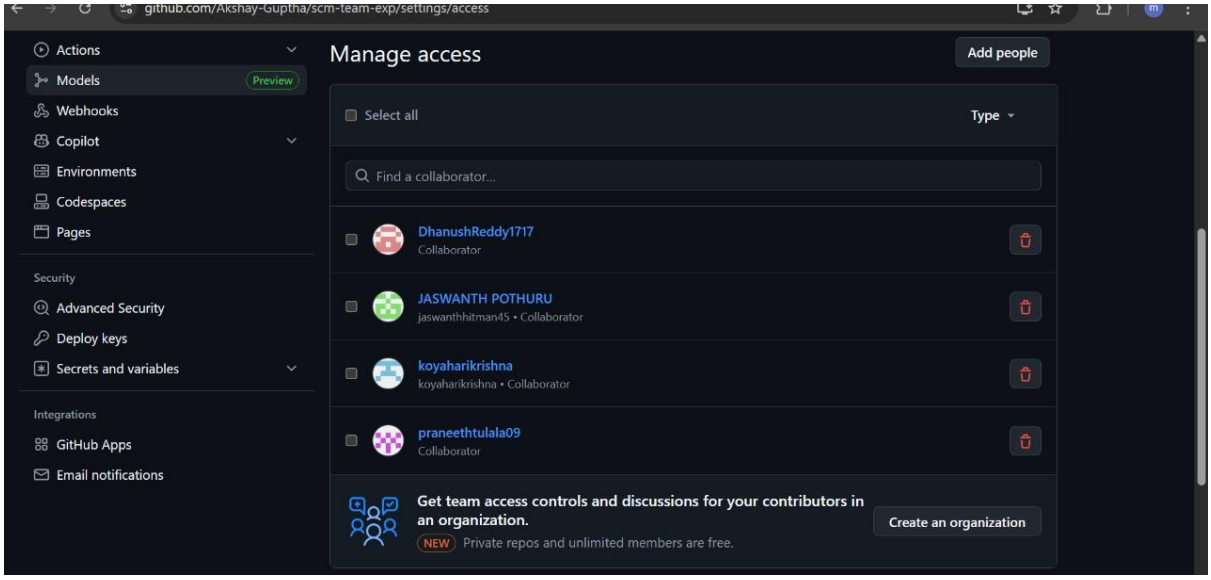


Step 9: Review and Merge by Owner

The repository owner reviewed the pull request and successfully merged it into the main branch.





Step 10: Final Verification

After merging, the main branch contained the base code along with the collaborator's contribution.